

# Project Title: Telecom Customer Churn Analysis

## Executive Summary

### Project Overview

The objective of this project was to analyze customer behavior and identify the major factors influencing customer churn in a telecom company. The analysis was performed using **Python**, **Pandas**, **NumPy**, **Matplotlib**, and **Seaborn** to clean, explore, and visualize customer data. The project focuses on uncovering actionable business insights that can help improve customer retention and reduce churn.

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### Dataset Summary

| Metric                   | Value      |
|--------------------------|------------|
| Total Records (Original) | 7,043      |
| Records After Cleaning   | 7,032      |
| Missing Values Removed   | 11 (0.16%) |
| Duplicate Customer IDs   | 0 (0%)     |
| Total Features           | 21         |

### Data Cleaning Performed

- Converted all column names to lowercase for consistency.
- Converted **TotalCharges** from object to numeric datatype.
- Removed **11 missing records (0.16%)**.
- Verified that **no duplicate customer IDs** existed.
- Standardized categorical values:
  - Senior Citizen (0 → No, 1 → Yes)
  - Multiple Lines ("No phone service" → "No")

This ensured a clean and reliable dataset for further analysis.

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# Key Business Insights

## 1. Customer Churn Rate

- 73.5% of customers were retained.
- 26.5% of customers churned.

### Business Insight

Although the majority of customers remain with the company, more than one out of every four customers leaves the service, indicating a significant opportunity to improve customer retention.

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## 2. Gender Distribution

The analysis showed that customer churn is almost equally distributed among male and female customers.

### Business Insight

Gender does **not** significantly influence customer churn and therefore should not be considered a major segmentation factor for retention strategies.

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## 3. Senior Citizens

- Senior Citizens: **1,142 customers (~16.2%)**
- Non-Senior Citizens: **5,890 customers (~83.8%)**

### Business Insight

The customer base is primarily composed of non-senior customers. Senior citizens represent a relatively small portion of the customer population.

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## 4. Customer Tenure Analysis

Customers with shorter tenure (especially within the first few months) showed the highest churn rates.

Customers with longer tenure demonstrated significantly higher retention.

### Business Insight

The first few months are the most critical stage of the customer lifecycle. Improving the onboarding experience and providing early engagement offers could substantially reduce churn.

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## 5. Contract Type Analysis

Month-to-Month customers experienced the highest churn.

Customers on One-Year and Two-Year contracts showed much lower churn rates.

### Business Insight

Long-term contracts increase customer loyalty and significantly reduce churn.

### Recommendation

- Promote annual subscription plans.
  - Offer discounts for long-term contracts.
  - Provide renewal incentives before contract expiration.
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## 6. Payment Method Analysis

Electronic Check emerged as the most commonly used payment method among customers.

The remaining payment methods had relatively similar adoption rates.

### Business Insight

Customers using Electronic Check should be monitored further, as this payment method is often associated with higher churn in telecom businesses.

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## 7. Internet Service Analysis

Customers using **Fiber Optic Internet** exhibited the highest churn.

### **Business Insight**

Despite offering faster internet speeds, Fiber Optic customers appear less satisfied, possibly due to higher pricing, service quality issues, or unmet expectations.

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## 8. Value-Added Services

Customers who subscribed to the following services showed noticeably lower churn:

- Online Security
- Device Protection
- Tech Support

Meanwhile,

- Phone Service
- Multiple Lines
- Streaming TV
- Streaming Movies

had relatively little influence on customer retention.

### **Business Insight**

Value-added protection and support services contribute significantly to customer loyalty.

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# Overall Business Findings

The analysis identifies several key drivers of customer churn:

- Short customer tenure
- Month-to-Month contracts
- Fiber Optic internet users
- Customers without Online Security
- Customers without Device Protection
- Customers without Tech Support

Conversely, customers enrolled in long-term contracts and additional support services demonstrate substantially higher retention rates.

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## Business Recommendations

### Customer Retention

- Target new customers during their first 3–6 months with personalized engagement campaigns.
- Encourage migration from Month-to-Month to annual or multi-year contracts.

### Service Improvement

- Investigate customer satisfaction among Fiber Optic users.
- Improve service quality and pricing strategies where necessary.

### Upselling Opportunities

Promote:

- Online Security
- Device Protection
- Tech Support

through bundled service packages to increase customer satisfaction and reduce churn.

### Customer Experience

Implement proactive customer support, loyalty rewards, and personalized retention campaigns for customers identified as high-risk.

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# Project Outcome

This project demonstrates the complete data analytics workflow, including:

- Data Cleaning and Preprocessing
- Exploratory Data Analysis (EDA)
- Data Visualization
- Business Insight Generation
- Strategic Recommendations
- Customer Churn Analysis

By transforming raw customer data into actionable business insights, the analysis provides practical recommendations that can help telecom companies improve customer retention, reduce churn, and enhance long-term customer value. The project showcases strong proficiency in **Python, Pandas, NumPy, Matplotlib, Seaborn, data preprocessing, exploratory data analysis, and business storytelling**, aligning with industry expectations for a Data Analyst portfolio project.